

Data Analytics for Personalized Genomics and Precision Medicine

Scribing of Lecture 7 – Clustering (Sep 30)

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In previous lecture (L6):

- Prof. Li recapped the importance of different data exploration and visualization techniques such as histogram and box plot.
- He lectured on motivation and ways of doing clustering.

In this lecture (L7):

- Prof. Li revised concepts related to **clustering**, **distance** and **similarity** as partially given in previous lecture.
- He provided more insights into **hierarchical clustering** and **Mahalanobis distance**.

Clustering (Partial Recap):

- Motivation of clustering:
 - For items: achieve better organization and faster searching when needed.
 - For people: fulfil different treatment/need for different groups.
 - For biology: identify co-expressed/differentially expressed genes, new disease subtypes and cell types.
- Clustering analysis: finding groups of objects such that the objects in a group will be similar to each other in the same group but different from ones in the other groups.
 - Result: intra-cluster differences are small while inter-cluster differences are large.
 - Components: data to be clustered, similarity measurement and clustering algorithm.

Distance (Partial Recap):

- Minkowski distance: a generalization of Euclidean distance with formulation of

$$\text{dist}(p, q) = \left(\sum_{k=1}^m |p_k - q_k|^r \right)^{\frac{1}{r}}$$

- Where r is a parameter, m is the number of dimensions (attributes), and p_k and q_k are the k -th attributes (components) of data objects p and q respectively.

- When $r = 1$: Manhattan or city-block distance, movement is restricted to gridlines, e.g. the number of bits that are different between two binary vectors.
- When $r = 2$: Euclidean distance (straight-line distance). Example:



Red - Manhattan distance ($r = 1$); Blue - Euclidean distance ($r = 2$)

- When r tends toward infinity: “supremum” distance, the maximum difference between any component of the vectors.
- Example of Minkowski distance:

point	x	y
p1	0	2
p2	2	0
p3	3	1
p4	5	1

L1	p1	p2	p3	p4
p1	0	4	4	6
p2	4	0	2	4
p3	4	2	0	2
p4	6	4	2	0

L2	p1	p2	p3	p4
p1	0	2.828	3.162	5.099
p2	2.828	0	1.414	3.162
p3	3.162	1.414	0	2
p4	5.099	3.162	2	0

L_{∞}	p1	p2	p3	p4
p1	0	2	3	5
p2	2	0	1	3
p3	3	1	0	2
p4	5	3	2	0

Distance Matrix

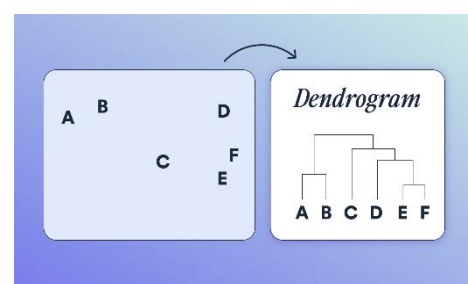
Hierarchical clustering:

➤ Meaning and purposes:

- Produces a set of nested clusters organized as a hierarchical tree
- Can be visualized as a dendrogram.
- May correspond to meaningful taxonomies, e.g. gene clusters, phylogeny reconstruction and animal kingdom.

➤ Steps:

- 1. Compute the similarity or distance matrix.
- 2. Let each data point be a cluster.



- 3. Merge the two closest clusters.
 - 4. Update the similarity or distance matrix.
 - 5. Repeat steps 3 and 4 until only a single cluster remains.
- Ways to update distance matrix: minimum, maximum, group average and distance between centroids.
- Example:

Gene	wt	mutant_1	mutant_2	mutant_3
At4g35770	1.5	3	3	1.5
At1g30720	4	7.5	7.5	5
At4g27450	1.5	1	1	1.5
At2g34930	10	25	23	15
At2g05540	1	1	2	1

- 1. Use Pearson correlation (shown below) to calculate the similarity matrix.

$$\rho_{X,Y} = \text{corr}(X, Y) = \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{E[(X - \mu_X)(Y - \mu_Y)]}{\sigma_X \sigma_Y}$$

Hence, we get

	At4g35770	At1g30720	At4g27450	At2g34930	At2g05540
At4g35770	1				
At1g30720	0.9733	1			
At4g27450	-1	-0.9733	1		
At2g34930	0.9493	0.9909	-0.9493	1	
At2g05540	0.5774	0.562	-0.5774	0.4528	1

- 2. Let each gene be a cluster and ignore the diagonal self-correlations.
- 3. Merge the two closest clusters (At1g30720 and At2g34930).

	At4g35770	At1g30720	At4g27450	At2g34930	At2g05540
At4g35770					
At1g30720	0.9733				
At4g27450	-1	-0.9733			
At2g34930	0.9493	0.9909	-0.9493		
At2g05540	0.5774	0.562	-0.5774	0.4528	

- 4. Update with minimum distance (largest correlation).

	At4g35770	At1g30720	At4g27450	At2g34930	At2g05540
At4g35770					
At1g30720	0.9733				
At4g27450	-1	->-0.9733			
At2g34930	0.9493	->0.9733	-0.9493		
At2g05540	0.5774	0.562	-0.5774	0.4528 ->0.562	

- 5. Merge the two closest clusters (At2g34930, At1g30720 and At4g35770 with 0.9733).

	At4g35770	At1g30720	At4g27450	At2g34930	At2g05540
At4g35770					
At1g30720	0.9733				
At4g27450	-1	-0.9493			
At2g34930	0.9733		-0.9493		
At2g05540	0.5774	0.562	-0.5774	0.562	

- 6. Update with minimum distance (largest correlation).

	At4g35770	At1g30720	At4g27450	At2g34930	At2g05540
At4g35770					
At1g30720					
At4g27450	-1	-0.9493			
At2g34930	->-0.9493	-0.9493			
At2g05540	0.5774	0.562	-0.5774	0.562	

- 7. Merge the two closest clusters (At2g34930, At1g30720, At4g35770, and At2g05540 with 0.5774).

	At4g35770	At1g30720	At4g27450	At2g34930	At2g05540
At4g35770					
At1g30720					
At4g27450	-0.9493	-0.9493			
At2g34930			-0.9493		
At2g05540	0.5774	0.5774	-0.5774	0.5774	

- 8. Update with minimum distance (largest correlation).

	At4g35770	At1g30720	At4g27450	At2g34930	At2g05540
At4g35770					
At1g30720					
At4g27450	-0.5774	-0.5774			
At2g34930			-0.5774		
At2g05540			-0.5774		

- 9. Merge the two closest clusters (At2g34930, At1g30720, At4g35770, At2g05540 and At4g27450 with -0.5774).

- 10. As there is now only a single cluster, the hierarchical clustering is completed.

➤ Resources to learn in Python:

- Scikit-learn: <https://scikit-learn.org/stable/>

- Hierarchical clustering: https://scikit-learn.org/stable/auto_examples/cluster/plot_agglomerative_dendrogram

- Potential project suggested by Prof. Li:
 - Data preprocessing for gene expression matrix:
 - ◆ Data collection and merging
 - ◆ Exploration
 - ◆ Visualization
 - ◆ Data cleaning
 - ◆ Distance matrix
 - ◆ **Clustering**

Mahalanobis distance:

- Motivation: calculate distance while also considering data distribution.
- Formulation:

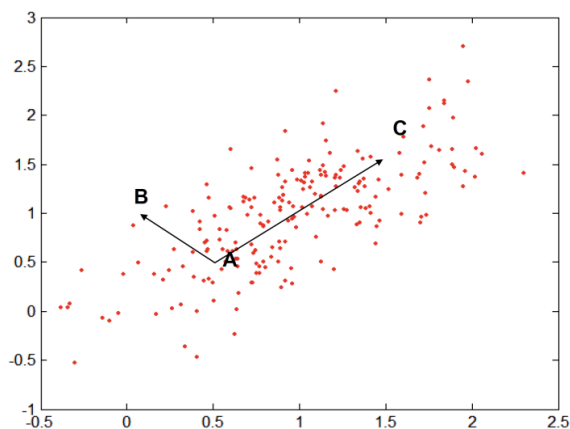
$$\text{mahalanobis}(p, q) = (p - q)^T \Sigma^{-1} (p - q)$$

- Where Σ is the covariance matrix that accounts for the variability of each feature as well as the relationship between features.
- To calculate the inverse of covariance matrix:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{\underset{\text{determinant}}{ad-bc}} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

- Example:

- 1. We have this graph and the covariance matrix:



Covariance matrix:

$$\Sigma = \begin{bmatrix} 0.3 & 0.2 \\ 0.2 & 0.3 \end{bmatrix}$$

- ◆ Where $A = [0.5, 0.5]$, $B = [0, 1]$ and $C = [1.5, 1.5]$
- ◆ Result: $\text{Mahal}(A, B) = 5$ and $\text{Mahal}(A, C) = 4$

- 2. Suppose we have two quizzes where in
 - ◆ Quiz 1: std = 10 and student A - student B = 1
 - ◆ Quiz 2: std = 1 and student C – student D = 1
 - ◆ Result: Two differences (A-B, C-D) need to be compared fairly, a way to do so is to consider the number of standard deviations away from two points.
- Comparison:
 - Min-max normalization: rescale each feature to a common range of [0,1].
 - Mahalanobis distance: accounts both feature variability and correlations.

Resources and uncovered topics:

- Introduction to Data Mining: Chapters 2.4 and 8
- K-means clustering
- Density-based clustering
- How to determine the number of clusters